

Chapter 06: Mechanical Properties of Metals

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Materials Science (MIAE 221)

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition) by William D. Callister, Jr. & David G. Rethwisch — Chapter 6

1. Executive Overview & First-Principles Philosophy

In engineering design, structural components are subjected to external mechanical forces during service: aircraft wing spars endure aerodynamic lift bending moments, automotive suspension control arms absorb road vibrations, and pressure vessels withstand internal fluid expansion.

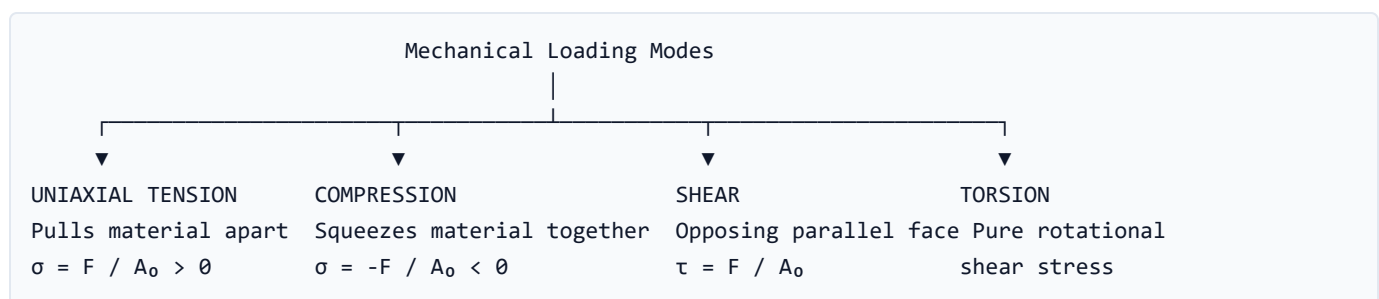
To ensure safety and prevent catastrophic collapse, mechanical engineers must quantify **how materials deform (elastically or plastically) and fracture under load**.

Mechanical properties are not arbitrary numbers; they are fundamental material characteristics evaluated through standardized laboratory testing (ASTM standards). The cornerstone test is the **Uniaxial Tensile Test**:

- It separates **reversible elastic deformation** (stretching of atomic bonds) from **permanent plastic deformation** (irreversible dislocation slip).
- It provides quantitative design parameters: stiffness (Young's modulus E), load capacity before permanent distortion (0.2 offset yield strength σ_y), maximum load carrying capability (ultimate tensile strength σ_u), energy absorption capacity (resilience U_r and toughness), and ductility (ϵ).

2. Concepts of Stress and Strain (Callister §6.2)

To compare materials of different sizes and cross-sections objectively, testing loads (F) and deformations (Δl) are normalized by original specimen geometry into **Stress (σ)** and **Strain (ϵ)**.



2.1 Engineering Stress & Engineering Strain

A. Engineering Stress (σ)

The instantaneous tensile force F applied perpendicular to the original cross-sectional area A_0 :

$$\sigma = \frac{F}{A_0}$$

- **SI Units:** Pascals ($\text{Pa} = \text{N/m}^2$). In engineering practice, we use Megapascals ($\text{MPa} = 10^6 \text{ N/m}^2 = \text{N/mm}^2$) or Gigapascals ($\text{GPa} = 10^9 \text{ N/m}^2 = 10^3 \text{ MPa}$).

B. Engineering Strain (ϵ)

The elongation of the specimen gauge length Δl divided by the initial gauge length l_0 :

$$\epsilon = \frac{l_i - l_0}{l_0} = \frac{\Delta l}{l_0}$$

- **Units:** Dimensionless (m/m or mm/mm). Frequently expressed as a percentage: .

C. Shear Stress (τ) & Shear Strain (γ)

- **Shear Stress (τ):** Force applied parallel to the planar cross-section:

$$\tau = \frac{F}{A_0}$$

- **Shear Strain (γ):** The tangent of the angular shear distortion angle θ (in radians):

$$\gamma = \tan \theta \approx \theta$$

3. Elastic Deformation & Hooke's Law (Callister §6.3 – §6.5)

3.1 Hooke's Law in Uniaxial Tension

When a metal is subjected to low stress levels, deformation is **elastic**: stress and strain are linearly proportional. If the load is released, the specimen snaps back instantly to its original dimensions with zero permanent distortion.

$$\sigma = E\epsilon$$

where E is the **Modulus of Elasticity (Young's Modulus)**.

- **Physical Origin:** E represents the macroscopic stiffness of the material—the slope of the linear elastic region. At the atomic scale, E is proportional to the curvature (second derivative) at the bottom of the interatomic potential energy well:

$$E \propto \left. \frac{d^2 E_N}{dr^2} \right|_{r_0}$$

- **Typical Modulus Values:**

- Diamond: $E \approx 1000 \text{ GPa}$
- Tungsten: $E \approx 407 \text{ GPa}$
- Structural Steel: $E \approx 207 \text{ GPa}$
- Titanium Alloys: $E \approx 107 - 115 \text{ GPa}$
- Aluminum Alloys: $E \approx 69 - 72 \text{ GPa}$
- Polyethylene (Polymer): $E \approx 0.2 - 1.0 \text{ GPa}$

3.2 Poisson's Ratio (ν)

When a solid is pulled in tension along the z -axis, it elongates axially ($\epsilon_z > 0$). Concurrently, to conserve volume, it **contracts laterally** in the transverse x and y directions ($\epsilon_x < 0, \epsilon_y < 0$).

Poisson's Ratio (ν) is defined as the negative ratio of lateral (transverse) strain to longitudinal (axial) strain:

$$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z} = -\frac{\Delta d/d_0}{\Delta l/l_0}$$

- **Theoretical Range:** For isotropic materials, $0 \leq \nu \leq 0.50$.
 - For ideal incompressible solids ($\Delta V = 0$): $\nu = 0.50$ (e.g., rubber).
 - For most engineering metals and alloys: $\nu \approx 0.25 - 0.35$ (typically $\nu \approx 0.33$).
 - For cork: $\nu \approx 0$ (does not expand laterally when compressed).

3.3 Elastic Relationships for Isotropic Materials

For an isotropic material (properties identical in all spatial directions), only two independent elastic constants exist. Young's modulus E , Shear modulus G , Bulk modulus K , and Poisson's ratio ν are mathematically linked:

$$G = \frac{E}{2(1 + \nu)}$$
$$K = \frac{E}{3(1 - 2\nu)}$$

- Since $\nu \approx 0.33$ for most metals:

$$G \approx \frac{E}{2(1 + 0.33)} = \frac{E}{2.66} \approx 0.38E \implies G \approx 0.4E$$

The shear modulus of a metal is roughly 40 of its tensile elastic modulus!

4. Tensile Properties & The Stress-Strain Curve (Callister §6.6)

yield point; stress subsequently rises with increasing strain. For metals that display this effect, the yield strength is taken as the average stress that is associated with the lower yield point because it is well defined and relatively insensitive to the testing procedure.¹⁰ Thus, it is not necessary to employ the strain offset method for these materials.

The magnitude of the yield strength for a metal is a measure of its resistance to plastic deformation. Yield strengths may range from 35 MPa (5000 psi) for a low-strength aluminum to greater than 1400 MPa (200,000 psi) for high-strength steels.

Concept Check 6.1 Cite the primary differences between elastic, anelastic, and plastic deformation behaviors.

The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

Tensile Strength

After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, point *M* in Figure 6.11, and then decreases to the eventual fracture, point *F*. The **tensile strength** *TS* (MPa or psi) is the stress at the maximum on the engineering stress-strain curve (Figure 6.11). This corresponds to the maximum stress that can be sustained by a structure in tension; if this stress is applied and maintained, fracture will result. All deformation to this point is uniform throughout the narrow region of the tensile specimen. However, at this maximum stress, a small constriction or neck begins to form at some point, and all subsequent deformation is confined at this neck, as indicated by the schematic specimen insets in Figure 6.11. This phenomenon is termed *necking*,

in lieu of *stress* because strength is a property of the metal, whereas stress is related to the magnitude of the force. In SI units, the unit of kilopounds per square inch (ksi) is sometimes used for the sake of convenience, where

Figure 6.11: Typical engineering stress-strain curve for a ductile metal showing elastic deformation, proportional limit *P*, 0.2 offset yield strength σ_y , ultimate tensile strength *M*, necking onset, and fracture point *F* — from Callister & Rethwisch 10th Ed. (Fig. 6.11).

4.1 Yielding & The 0.2 Offset Yield Strength (σ_y)

As stress increases beyond the elastic limit, atomic planes begin to slide over one another via dislocation motion. The material undergoes **plastic deformation** (permanent, unrecoverable strain).

Because the transition from elastic to plastic behavior is gradual, engineers use the **0.2 Strain Offset Method** to define a reproducible yield strength:

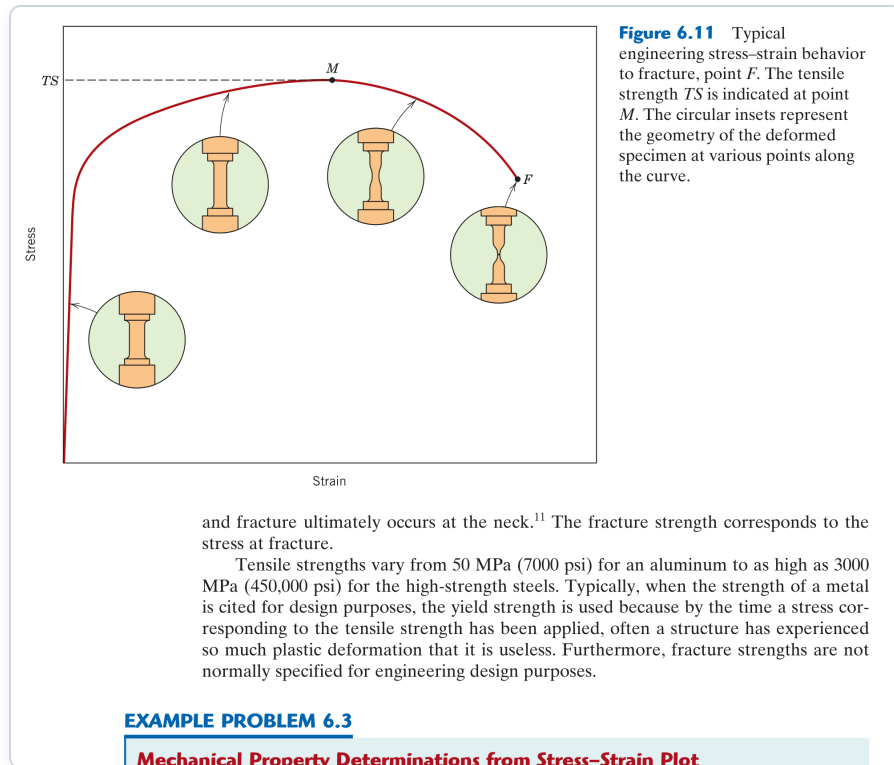


Figure 6.12: Determination of the 0.2 offset yield strength σ_y on a brass alloy stress-strain curve by drawing a line parallel to the elastic modulus starting at $\epsilon = 0.002$ — from Callister & Rethwisch 10th Ed. (Fig. 6.12).

The 3-Step Offset Yield Algorithm:

1. Locate strain $\epsilon = 0.002$ (0.2) on the horizontal strain axis.
2. Construct a straight line starting at $\epsilon = 0.002$ with a slope exactly equal to the initial Young's modulus (E).
3. The stress at the intersection of this parallel line with the experimental stress-strain curve is the **0.2 Offset Yield Strength** (σ_y).

- **Yield Point Phenomenon in Low-Carbon Steels:** Certain annealed steels display an abrupt transition: an **Upper Yield Point** followed by an immediate drop to a **Lower Yield Point** where plastic flow propagates at constant stress (Lüders bands).

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Physical Cause

: Small interstitial carbon and nitrogen atoms diffuse to the core of edge dislocations, forming dense **Cottrell atmospheres** that lock the dislocations. Once the stress reaches the upper yield point, dislocations break free from their carbon clouds and multiply rapidly, lowering the stress required to sustain plastic flow.

4.2 Ultimate Tensile Strength (UTS / σ_u) & Necking

The **Ultimate Tensile Strength** (σ_u) is the maximum engineering stress recorded on the engineering stress-strain curve:

$$\sigma_u = \frac{F_{\max}}{A_0}$$

- **Deformation Modes Before and After UTS:**

- **From $\sigma = 0$ up to σ_u :** Plastic deformation is **uniform** along the entire gauge length. Strain hardening strengthens the metal faster than the cross-sectional area thins.
 - **At $\sigma = \sigma_u$:** Strain hardening can no longer compensate for cross-sectional thinning. A localized geometric instability occurs: **Necking initiates**.
 - **From σ_u to Fracture:** All subsequent plastic deformation concentrates exclusively within the necked region. The engineering stress falsely appears to decrease until the specimen tears apart at the fracture stress σ_f .
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4.3 Ductility: and

Ductility measures the degree of plastic deformation a material can sustain before fracture:

1. **Percent Elongation ():**

where l_f is the fracture gauge length measured after fitting the broken pieces back together.

2. **Percent Reduction in Area ():**

where A_f is the minimum cross-sectional area measured at the fracture neck.

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Engineering Threshold

: Materials with are classified as **ductile**; materials with are classified as **brittle**.

4.4 Modulus of Resilience (U_r) & Modulus of Toughness

A. Modulus of Resilience (U_r)

Resilience is the capacity of a material to absorb energy when deformed elastically and release that energy upon unloading without permanent distortion. Mathematically, U_r is the area under the engineering stress-strain curve up to yielding:

$$U_r = \int_0^{\epsilon_y} \sigma, d\epsilon = \frac{1}{2} \sigma_y \epsilon_y$$

Using Hooke's Law $\epsilon_y = \frac{\sigma_y}{E}$:

$$U_r = \frac{\sigma_y^2}{2E}$$

- **SI Units:** $\text{J/m}^3 = \text{Pa} = \text{N} \cdot \text{m/m}^3$.
- **Engineering Design:** High resilience requires a **high yield strength** (σ_y) combined with a **low elastic modulus** (E) (e.g., mechanical spring steels, archery bows).

B. Modulus of Toughness

Toughness is the total mechanical energy absorbed per unit volume up to the point of catastrophic fracture.

- Mathematically: The **total area under the entire engineering stress-strain curve** from $\epsilon = 0$ to ϵ_f .
- **Requirement for High Toughness:** A material must exhibit **BOTH high strength AND high ductility**. A ceramic has high strength but zero ductility $\implies$ low toughness. Pure lead has extreme ductility but near-zero strength $\implies$ low toughness. Structural alloy steels possess high strength and high ductility $\implies$ exceptionally high toughness.

5. True Stress and True Strain (Callister §6.7)

Why does the engineering stress-strain curve show a drop in stress after necking ($\sigma_u \rightarrow \sigma_f$)? Because engineering stress $\sigma = F/A_0$ divides the load by the **initial area** A_0 , which ignores the dramatic thinning occurring in the neck! The actual load-bearing area A_i decreases far faster than the load decreases, meaning the material inside the neck is actually becoming stronger right up to fracture!

To represent the true physical behavior of the metal, we define:

1. **True Stress** (σ_T): Instantaneous load divided by instantaneous cross-sectional area:

$$\sigma_T = \frac{F}{A_i}$$

2. **True Strain** (ϵ_T): Incremental logarithmic strain:

$$\epsilon_T = \int_{l_0}^{l_i} \frac{dl}{l} = \ln \left(\frac{l_i}{l_0} \right)$$

5.1 Conversion Formulas (Valid ONLY up to Necking Onset)

Assuming plastic deformation occurs at **constant volume** ($V = A_0 l_0 = A_i l_i \implies \frac{A_0}{A_i} = \frac{l_i}{l_0}$):

$$\begin{aligned} \frac{l_i}{l_0} &= \frac{l_0 + \Delta l}{l_0} = 1 + \epsilon \\ \sigma_T &= \frac{F}{A_i} = \frac{F}{A_0} \left(\frac{A_0}{A_i} \right) = \sigma \left(\frac{l_i}{l_0} \right) = \sigma(1 + \epsilon) \\ \epsilon_T &= \ln \left(\frac{l_i}{l_0} \right) = \ln(1 + \epsilon) \end{aligned}$$

- **⚠ Critical Boundary Condition:** These formulas are strictly valid **only up to the onset of necking** ($\sigma \leq \sigma_u$). Once necking initiates, deformation is non-uniform, and true stress must be calculated directly from measured neck geometry: $\sigma_T = F/A_{\text{neck}}$ and $\epsilon_T = \ln(A_0/A_{\text{neck}})$.

5.2 Strain Hardening & The Hollomon Power Law

For most metals in the plastic regime between yielding and necking, the true stress-true strain curve obeys the **Hollomon Equation**:

$$\sigma_T = K \epsilon_T^n$$

where:

- **K = Strength Coefficient** (MPa).
- **n = Strain-Hardening Exponent** ($0 \leq n \leq 1$). High n means the material hardens rapidly with deformation.
- **Necking Criterion (Considère's Construction)**: Necking begins at the precise point where the true strain equals the strain-hardening exponent:

$$\epsilon_{T,\text{neck}} = n$$

6. Hardness Testing (Callister §6.10)

Hardness is a measure of a material's resistance to localized surface plastic indentation (scratching, piercing, or denting).

Hardness Test	Indenter Geometry	Applied Load	Measurement Metric
Rockwell (HRA, HRB, HRC)	Diamond cone (120°) or hardened steel sphere ($1/16''$)	Minor load (10 kg) + Major load (60 – 150 kg)	Depth of permanent indentation read directly on dial. Fast, non-destructive.
Brinell (HB)	10 mm tungsten carbide sphere	500 – 3000 kg for 10 – 15 s	Diameter of surface impression (d) measured under optical microscope.
Vickers (HV)	Square-base diamond pyramid (136°)	1 – 120 kg	Diagonal impression lengths d_1, d_2 . Microhardness of thin coatings.
Knoop (HK)	Elongated rhombohedral diamond pyramid	Micro-loads (10 – 1000 g)	Major diagonal length. Ideal for brittle ceramics and glass.

Correlation Between Hardness and Tensile Strength (for Steels)

For most structural steels, Brinell hardness (**HB**) correlates directly with Ultimate Tensile Strength:

$$\text{UTS (MPa)} \approx 3.45 \times \text{HB}$$

$$\text{UTS (psi)} \approx 500 \times \text{HB}$$

7. Comprehensive Step-by-Step Problem Walkthroughs

7.1 Problem 1: Full Tensile Data Reduction of an Aerospace Alloy

Problem Statement: A cylindrical tensile specimen of an aluminum alloy has an original diameter $d_0 = 12.8 \text{ mm}$ and original gauge length $l_0 = 50.800 \text{ mm}$. A tensile test produces the following mechanical data:

- Elastic linear slope: at load $F = 35.0 \text{ kN}$, gauge length is $l = 50.985 \text{ mm}$.
- 0.2 strain offset load: $F_y = 51.0 \text{ kN}$.
- Maximum load: $F_{\max} = 72.5 \text{ kN}$ at gauge length $l = 56.400 \text{ mm}$.
- Fracture load: $F_f = 61.0 \text{ kN}$ at fracture length $l_f = 60.200 \text{ mm}$ and fracture diameter $d_f = 10.4 \text{ mm}$.

Calculate:

1. Young's modulus (E).
2. 0.2 offset yield strength (σ_y).
3. Ultimate tensile strength (σ_u).
4. Percent elongation () and percent reduction in area ().
5. Modulus of resilience (U_r).
6. True stress and true strain at the onset of necking (maximum load).

Step 1: Calculate Original Cross-Sectional Area A_0

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi (12.8 \text{ mm})^2}{4} = 128.68 \text{ mm}^2 = 1.2868 \times 10^{-4} \text{ m}^2$$

Step 2: Calculate Young's Modulus (E)

At $F = 35.0 \text{ kN} = 35,000 \text{ N}$:

$$\sigma = \frac{35,000 \text{ N}}{1.2868 \times 10^{-4} \text{ m}^2} = 2.720 \times 10^8 \text{ Pa} = 272.0 \text{ MPa}$$

$$\epsilon = \frac{l - l_0}{l_0} = \frac{50.985 - 50.800}{50.800} = \frac{0.185 \text{ mm}}{50.800 \text{ mm}} = 0.003642$$

$$E = \frac{\sigma}{\epsilon} = \frac{272.0 \text{ MPa}}{0.003642} = 74,680 \text{ MPa} \approx 74.7 \text{ GPa}$$

Step 3: Calculate 0.2 Offset Yield Strength (σ_y)

$$\sigma_y = \frac{F_y}{A_0} = \frac{51,000 \text{ N}}{1.2868 \times 10^{-4} \text{ m}^2} = 3.963 \times 10^8 \text{ Pa} = 396.3 \text{ MPa} \approx 396 \text{ MPa}$$

Step 4: Calculate Ultimate Tensile Strength (σ_u)

$$\sigma_u = \frac{F_{\max}}{A_0} = \frac{72,500 \text{ N}}{1.2868 \times 10^{-4} \text{ m}^2} = 5.634 \times 10^8 \text{ Pa} = 563.4 \text{ MPa} \approx 563 \text{ MPa}$$

Step 5: Calculate Ductility (and)

- Percent Elongation:
- Final Neck Area A_f :

$$A_f = \frac{\pi d_f^2}{4} = \frac{\pi (10.4 \text{ mm})^2}{4} = 84.95 \text{ mm}^2$$

- Percent Reduction in Area:

Step 6: Calculate Modulus of Resilience (U_r)

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(396.3 \times 10^6 \text{ Pa})^2}{2 \times (74.7 \times 10^9 \text{ Pa})} = \frac{1.5705 \times 10^{17}}{1.494 \times 10^{11}} = 1.051 \times 10^6 \text{ J/m}^3 = 1.05 \text{ MJ/m}^3$$

Step 7: Calculate True Stress and True Strain at Necking Onset

At maximum load ($l = 56.400 \text{ mm}$), necking is just beginning, so conversion formulas are valid:

$$\epsilon = \frac{56.400 - 50.800}{50.800} = \frac{5.600}{50.800} = 0.1102$$
$$\sigma = \sigma_u = 563.4 \text{ MPa}$$

- True Strain:

$$\epsilon_T = \ln(1 + \epsilon) = \ln(1 + 0.1102) = \ln(1.1102) = 0.1045$$

- True Stress:

$$\sigma_T = \sigma(1 + \epsilon) = 563.4 \times (1 + 0.1102) = 625.5 \text{ MPa}$$

- Notice that at necking, true stress (625.5 MPa) is significantly higher than engineering UTS (563.4 MPa) due to cross-sectional reduction!

7.2 Problem 2: Lateral Contraction via Poisson's Ratio

Problem Statement: A cylindrical rod of titanium alloy ($E = 107 \text{ GPa}$, $\nu = 0.34$) with initial diameter $d_0 = 10.000 \text{ mm}$ is subjected to a tensile load of $F = 25.0 \text{ kN}$.

1. Verify whether deformation is strictly elastic if the yield strength is $\sigma_y = 825 \text{ MPa}$.
2. Calculate the change in diameter Δd of the rod under this load.

Step 1: Check Elastic Condition

$$A_0 = \frac{\pi (0.010 \text{ m})^2}{4} = 7.854 \times 10^{-5} \text{ m}^2$$

$$\sigma = \frac{25,000 \text{ N}}{7.854 \times 10^{-5} \text{ m}^2} = 3.183 \times 10^8 \text{ Pa} = 318.3 \text{ MPa}$$

Since $\sigma = 318.3 \text{ MPa} < \sigma_y = 825 \text{ MPa}$, deformation is **strictly elastic**. Hooke's Law and Poisson's ratio apply!

Step 2: Calculate Axial Strain ϵ_z

$$\epsilon_z = \frac{\sigma}{E} = \frac{318.3 \times 10^6 \text{ Pa}}{107 \times 10^9 \text{ Pa}} = 2.975 \times 10^{-3} = 0.002975$$

Step 3: Calculate Lateral Strain ϵ_x via Poisson's Ratio

$$\epsilon_x = -\nu\epsilon_z = -(0.34) \times (0.002975) = -1.0115 \times 10^{-3}$$

Step 4: Calculate Change in Diameter Δd

$$\epsilon_x = \frac{\Delta d}{d_0} \implies \Delta d = \epsilon_x \cdot d_0 = (-1.0115 \times 10^{-3}) \times (10.000 \text{ mm}) = -0.0101 \text{ mm} = -10.1 \mu\text{m}$$

Conclusion

: Under the 25 kN tensile load, the titanium rod diameter contracts by 10.1 μm .

8. Critical Concordia Exam Traps & Red Flags

- **⚠ Trap 1: Applying $\sigma_T = \sigma(1 + \epsilon)$ Beyond Necking:** The equations $\sigma_T = \sigma(1 + \epsilon)$ and $\epsilon_T = \ln(1 + \epsilon)$ are based on uniform elongation and constant volume. **They are invalid beyond the maximum load (UTS)!** If an exam question asks for true stress at fracture, you **must** use $\sigma_{T,f} = F_f/A_f$, NOT $\sigma_f(1 + \epsilon_f)$!
- **⚠ Trap 2: Forgetting to Convert GPa to MPa:** In resilience $U_r = \frac{\sigma_y^2}{2E}$, students often square σ_y in MPa and divide by E in GPa without converting units.
 - Correct practice: Convert everything to pure SI units (Pa): σ_y in Pa, E in Pa $\implies U_r$ in $\text{J/m}^3 = \text{Pa}$.
- **⚠ Trap 3: Area Calculation with Diameter vs. Radius:** Area is $A = \frac{\pi d^2}{4}$, NOT πd^2 . A missing factor of 4 causes stress to be underestimated by 400!
- **⚠ Trap 4: Negative Sign in Poisson's Ratio:** Remember that Δd is negative in tension (the rod gets thinner). A lateral strain $\epsilon_x = -\nu\epsilon_z$ is negative. Always report $\Delta d = -10.1 \mu\text{m}$ (or explicitly state "a diameter reduction of 10.1 μm ").